

IntelliCane

Smart Assistive Cane with Haptic Obstacle Avoidance, Fall Detection, and a Mobile App for Blind Users and their Caretakers

Team Neural-Nexus | Track A: Embedded IoT System Development | IoTrix 2.0

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Repository: <https://github.com/Neural-Nexus-IoTrix-2-0/intellicane> | Submission Date: 5 October 2026

1. Problem Definition & Intended Use Case

Visually impaired individuals encounter hazardous obstacles during daily mobility, including low-lying clutter, knee-to-waist hazards, and sudden vertical drops. Furthermore, accidental cane drops or user falls can leave individuals stranded without immediate assistance. Existing commercial Electronic Travel Aids (ETAs) present three critical shortcomings: (1) prohibitive acquisition costs often exceeding \$500 USD (LKR 150,000+), (2) architectural reliance on cloud connectivity for basic hazard detection, and (3) acoustic audio spam that obstructs natural hearing.

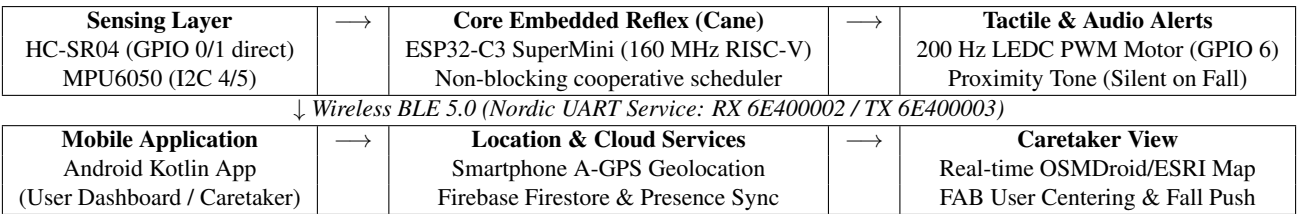
IntelliCane addresses these issues directly. It provides an ultra-low-cost (LKR 2,800 bench electronics), 100% offline-safe assistive white cane that pairs real-time haptic obstacle guidance and inertial fall detection with a dedicated companion Android mobile application for blind users and their caretakers.

2. Proposed Solution & System Overview

The IntelliCane ecosystem integrates an intelligent embedded controller on the cane with a dual-mode companion smartphone application:

- Safety-Critical Cane Reflex (ESP32-C3):** Executes a sub-50 ms deterministic loop. An HC-SR04 ultrasonic transceiver detects forward hazards up to 4 meters, driving a 3-pin vibration motor via 200 Hz hardware LEDC PWM that scales smoothly between 60 cm and 10 cm. A 6-axis MPU6050 IMU detects orientation and impact forces ($> 65^\circ$ tilt or $> 2.5g$). Upon detecting a fall, the cane automatically enters a silent emergency state (disabling local buzzer and motor to avoid user distress) while immediately routing high-priority alert packets over BLE.
- Companion Android Application (com.example.intellicane):** Connects wirelessly over Bluetooth Low Energy (BLE 5.0 Nordic UART Service). Operates in two dedicated modes:
 - Blind User Dashboard:* Accessible interface with connection indicators, one-tap A-GPS location sharing, and a collapsible floating terminal window for on-demand telemetry diagnostics without visual clutter.
 - Caretaker Mode:* Real-time presence tracking, automated emergency fall alerts synchronized directly via Firebase Firestore, and an interactive ESRI/OpenStreetMap (OSMDroid) interface with an instant user-centering FAB (18.5x zoom)—eliminating the cost, weight, and power draw of a standalone hardware GPS module.

3. System Architecture & Data Flow



4. Hardware Implementation & Circuit Design

The cane hardware is designed around the stamp-sized **ESP32-C3 SuperMini** powered by a regulated 5 V rail:

- Ultrasonic Ranging (HC-SR04):** Triggered via 10 μ s pulses on GPIO 0. The Echo pulse is connected directly to GPIO 1, keeping the bench prototype wiring minimal, robust, and efficient.
- Inertial Motion Sensor (GY-521 MPU6050):** Communicates over hardware I2C (SDA: GPIO 4, SCL: GPIO 5) at 100 kHz. AD0 is grounded to set address 0x68. A custom direct-register fallback driver ensures resilience against clone ICs and bus lockups.
- Tactile Haptic Motor Module:** 3-pin vibration motor module with onboard MOSFET driver powered from 5 V; gate modulated via GPIO 6 at 200 Hz PWM, providing gradual spatial perception without hand numbness.
- Audible Hazard Alarm:** Active/passive 3.3 V buzzer on GPIO 7 provides urgent pulsed alerts for close obstacles (< 30 cm), while automatically silencing during fall events to prevent user panic.

5. Firmware Architecture & Control Logic

The firmware (firmware/core-sensing/core-sensing.ino) operates with zero blocking delay() calls, utilizing a cooperative time-sliced scheduler:

- Task Rates:** Ultrasonic ranging @ 20 Hz (50 ms), IMU tilt tracking @ 50 Hz (20 ms), Actuator PWM updates @ 100 Hz (10 ms), and BLE telemetry transmission @ 10 Hz (100 ms).
- Proportional Ramping Equation:** The PWM duty cycle ramps monotonically as obstacle distance d decreases:

$$\text{PWM Duty} = \text{constrain} \left(255 - \frac{d - d_{\min}}{d_{\max} - d_{\min}} \times (255 - \text{PWM}_{\min}), \text{PWM}_{\min}, 255 \right) \quad (d_{\min} = 10\text{cm}, d_{\max} = 60\text{cm})$$

- **Silent Fall Detection & BLE Dispatch:** Tilt > 65° or impact acceleration > 2.5g triggers emergency fall mode. The cane silences local haptic and audible alerts, transitions the onboard LED, and immediately broadcasts FALL_STATE:1 and [ALERT: FALL DETECTED!] over BLE. Restoring the cane upright (< 30°) transmits FALL_STATE:0 and automatically restores normal navigation.

6. Companion Mobile Application (application/)

The native Android application is structured in Kotlin (com.example.intellicane) to bridge IoT telemetry:

- **BLE GATT Central Client:** Auto-discovers Intelligent-Cane, connects to Nordic UART Service, and parses telemetry: DIST:...|TILT:...|FALL:...|MOTOR:...|BUZZ:...|BLE:....
- **User Dashboard & Floating Terminal:** Blind users view simple high-contrast status and location-sharing controls. A collapsible floating terminal toggle (btnToggleTerminal) allows on-demand viewing of live sensor data without cluttering the screen.
- **Caretaker Map & Emergency Centering:** Integrates org.osmdroid with ESRI tile sources. Features a dedicated FAB (fabUserLocation) to immediately center and zoom (18.5x) on the blind user’s real-time A-GPS coordinates.
- **Firebase Cloud Backend:** Syncs state with Cloud Firestore and Realtime Database. Upon receiving FALL_STATE:1, updateFallStateInFirestore immediately flags the emergency in the cloud to alert caretakers.

7. Testing, Validation & Verification Evidence

- **Host Unit Testing** (tests/proximity_feedback_test.cpp): 100% pass across boundary conditions, invalid inputs (NaN, ∞, negative distances), monotonic PWM progression, motor polarity inversion, and 32-bit millis() rollover resilience.
- **Bench Distance Calibration:** HC-SR04 verified against physical tape measurements from 2 cm to 250 cm with ±1.5 cm accuracy. Motor PWM engages deterministically at 59.9 cm and hits 100% duty at 10 cm.
- **Tilt Verification:** Calibrated with a digital protractor; fall alarm triggers reliably at > 65° and silences within 200 ms of cane recovery (< 30°).
- **Wireless BLE Performance:** Telemetry stream validated at 10 Hz to smartphone with < 25 ms latency and up to 10 m range.

8. Budget & Cost Feasibility

The prototype achieves industry-leading cost efficiency by leveraging the user’s smartphone for GPS and connectivity:

Component	Specification / Purpose	Phase 1 Cost (LKR)		Status
ESP32-C3 SuperMini	RISC-V 160 MHz MCU with BLE 5.0 & Na-tive USB-C	1,400		Deployed & Verified
HC-SR04 Transceiver	Forward ultrasonic obstacle ranging (2–400 cm)	450		Deployed & Verified
GY-521 MPU6050	6-Axis Accelerometer / Gyroscope for tilt & fall	600		Deployed & Verified
3-Pin Vibration Motor	Integrated driver module (200 Hz LEDC PWM)	250		Deployed & Verified
Piezo Buzzer & Interconnects	High-urgency audio hazard tone & bench wiring jumpers	100		Deployed & Verified
Total Electronics Cost	Safety-Critical Bench Prototype	LKR 2,800	Budget Ceiling: LKR 35,000	

9. Limitations, Risks & Mitigations

- **Ultrasonic Specular Scattering:** Sound waves can scatter on angled soft fabrics beyond 2 meters. *Mitigation:* For the final build, a Time-of-Flight (ToF) laser ranging sensor will be integrated as a complementary secondary channel.
- **Breadboard Jumper Durability:** Prototype jumpers can loosen under prolonged mechanical tapping. *Mitigation:* Migrating to a custom two-layer PCB and vibration-isolated enclosure before the final competition.
- **Smartphone Disconnection:** If the phone battery depletes, remote caregiver tracking stops. *Mitigation:* Core obstacle avoidance and local audio-visual fall alarms remain 100% operational offline on the cane.

10. Roadmap Prior to the Final (17 October 2026)

1. **Hardware Enclosure & Custom PCB:** Fabricate a custom PCB and 3D-print an ergonomic cane handle housing with TPU shock damping.
2. **End-to-End Caretaker Alert Flow:** Finalize Firebase Cloud Functions to dispatch automated SMS and push notifications upon verified fall events.
3. **Battery Optimization:** Integrate an 18650 Li-ion cell with TP4056 charging circuit and configure ESP32-C3 Light Sleep for > 24 hour operation.

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